

Report on Drainage Design for Open Field of the Civil Engineering Department Field at Tezpur University.

Project title: Design and Analysis of an Open Field Drainage System
of the Civil Engineering Department field at Tezpur University,
Including Runoff Estimation and Component Sizing Based on
Site-Specific Data.



Submitted By:

CEB22035
CEB22036
CEB22037
CEB22038
CEB22039
CEB22040

CEB22041
CEB22042
CEB22043
CEB22044
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CEB22053

ABSTRACT

Our report presents the design process and analysis of an open-field drainage system intended to address waterlogging in the Civil engineering department field. The objective of the drainage system is to improve soil aeration and reduce water stagnation. The report outlines the methodology for determining the required drainage capacity and the layout of the drainage system. The analysis includes soil permeability, rainfall data, and field topography to design an efficient system.

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1.INTRODUCTION

Drainage design is a crucial aspect of infrastructure planning, particularly in areas where water management is essential to prevent waterlogging and structural damage. At Tezpur University's Civil Engineering Department, the surrounding open field requires an effective drainage system to manage rainfall and runoff efficiently.

The purpose of this project is to design a drainage system tailored to the unique topography, soil characteristics, and rainfall patterns of the Civil engineering department field. The project focuses on developing an open field drainage layout that ensures safe and secure drainage, while preserving the integrity of the field and surrounding structures.

Key objectives include:

- Estimating runoff volumes based on local climatic data.
Designing an open channel to control and direct runoff away from vulnerable areas.
- Incorporating erosion control measures to maintain soil stability and minimize maintenance.

This report presents the findings from a site analysis, including topography, followed by calculations of runoff and design specifications for various drainage components. The final design aims to achieve a sustainable, low-maintenance solution that effectively addresses water management needs for the Civil Engineering department's field area.

2.THEORY AND BACKGROUND

An open-field drainage system is designed to efficiently manage surface water and promote groundwater infiltration, helping to prevent waterlogging and maintain soil health. Such systems typically consist of a network of ditches, swales or trenches that channel excess water away from key areas. The design of these systems is guided by several critical factors:

- **Rainfall Intensity:** The local rainfall intensity impacts the volume of runoff that the system must handle, guiding calculations for channel dimensions and layout.
- **Topography:** Land slope and natural contours dictate water flow direction, helping to optimize drainage system placement and ensure effective water movement.
- **Drainage Coefficient:** This is a measure of the system's drainage capacity, typically expressed in mm/day, representing the volume of water that the system can effectively drain.

The design process involves calculating the necessary drainage capacity based on local climate data, soil conditions, and environmental requirements of the field area.

3. SITE MAP

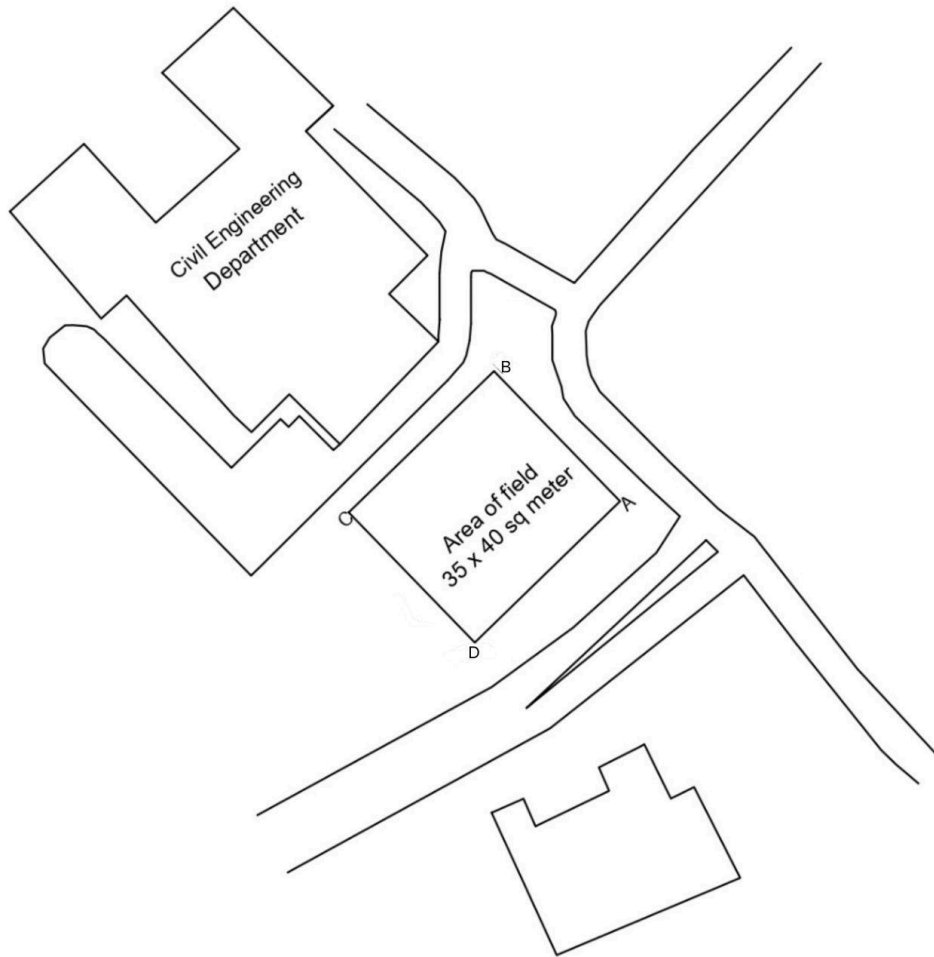


Fig: Layout map of Department of Civil Engineering Field.

4.METHODOLOGY

4.1. Drainage Design Parameters:

- Field Area: 1400 m²
- Runoff coefficient = 0.6 for open field
- Intensity of rain = 90 mm/hr

From rational method

$Q = (\text{area of ground}) \times (\text{runoff coefficient}) \times (\text{intensity of rainfall})$
 where Q is peak runoff rate.

Manning's equation

$$Q = (1/n)AR^{2/3}S^{1/2}$$

4.2 Hydrological Data:

- Runoff is expected to be high during the monsoon season, necessitating efficient drainage.

5.DESIGN CALCULATIONS

To determine the required drainage system design, the following steps were followed:

1. Calculating Drainage Requirements:

- Using the formula for peak runoff rate by using rational method

From rational method

$$Q = (C \times i \times A)$$

Where Q is peak runoff rate (m^3/hr)

C is runoff coefficient

i is rainfall intensity (mm/hr)

A is drainage area (m^2)

For the Civil engineering department field -

$$Q = 0.6 \times 90 \times 1400$$

$$Q = 75.6 \text{ m}^3/\text{hr}$$

$$Q = 21 \times 10^{-3} \text{ m}^3/\text{s}$$

$$Q = 21 \text{ L/s}$$

If the rainfall occurs with maximum intensity for 2 hours continuously the water collected will be equal = 2×75.6

$$= 151.2 \text{ m}^3$$

For most economical cases bed slope = $1/1000$

Design of drain AB

Assuming 15% of the runoff goes into AB drain

$$Q = 15/100 \times 21 \text{ L/s}$$

$$= 3.15 \text{ L/s}$$

Design of drain BC

Assuming 10% of the runoff goes into BC drain

$$Q = 10/100 \times 21 \text{ L/s}$$

$$= 2.1 \text{ L/s}$$

Design of drain AD

Assuming 85% of the runoff goes into AD drain

$$Q = 85/100 \times 21 \text{ L/s}$$

$$= 17.85 \text{ L/s}$$

For rectangular channel

$$P = B + 2Y$$

Where P is perimeter,
 B is width,
 Y is depth.

$$A = BY$$

Where A is area ,
 B is width,
 Y is depth.

From Manning's equation

$$Q = (1/n)AR^{2/3}S^{1/2}$$

For drain AB

$$3.15 \times 10^3 = 1/0.015 \times BY \times \{(BY)/(B+2Y)\}^{2/3} \times 1/1000^{1/2}$$

$$Y^{8/3} = 3.15 \times 10^{-3} \times 1000^{1/2} \times 2^{2/3} \times 0.015 / 2$$

$$Y = 8 \text{ cm}$$

For drain BC

$$2.1 \times 10^3 = 1/0.015 \times BY \times \{(BY)/(B+2Y)\}^{2/3} \times 1/1000^{1/2}$$

$$Y^{8/3} = 2.1 \times 10^{-3} \times 1000^{1/2} \times 2^{2/3} \times 0.015 / 2$$

$$Y = 7 \text{ cm}$$

For drain AD

$$17.85 \times 10^3 = 1/0.015 \times BY \times \{(BY)/(B+2Y)\}^{2/3} \times 1/1000^{1/2}$$

$$Y^{8/3} = 17.85 \times 10^{-3} \times 1000^{1/2} \times 2^{2/3} \times 0.015 / 2$$

$$Y = 16 \text{ cm}$$

For most economical case

$$B=2Y$$

For drainage AB: B = 16 cm

For drainage BC: B = 14 cm

For drainage AD: B = 30 cm

NOTE: For our ease of calculation, we have taken the width the same, i.e., 25 cm and according to that our depth changes.

So the new depth are -

For AB Drain - 12 cm

For AD Drain - 30 cm

For BC Drain - 10 cm

DRAIN DIMENSION

The open drains were designed with a rectangular cross-section:

1) For AB drain

- Bottom width: 25cm
- Bed slopes: 1:1000 (vertical:horizontal)
- Depth:12cm

2) For AD drain

Bottom width: 25cm

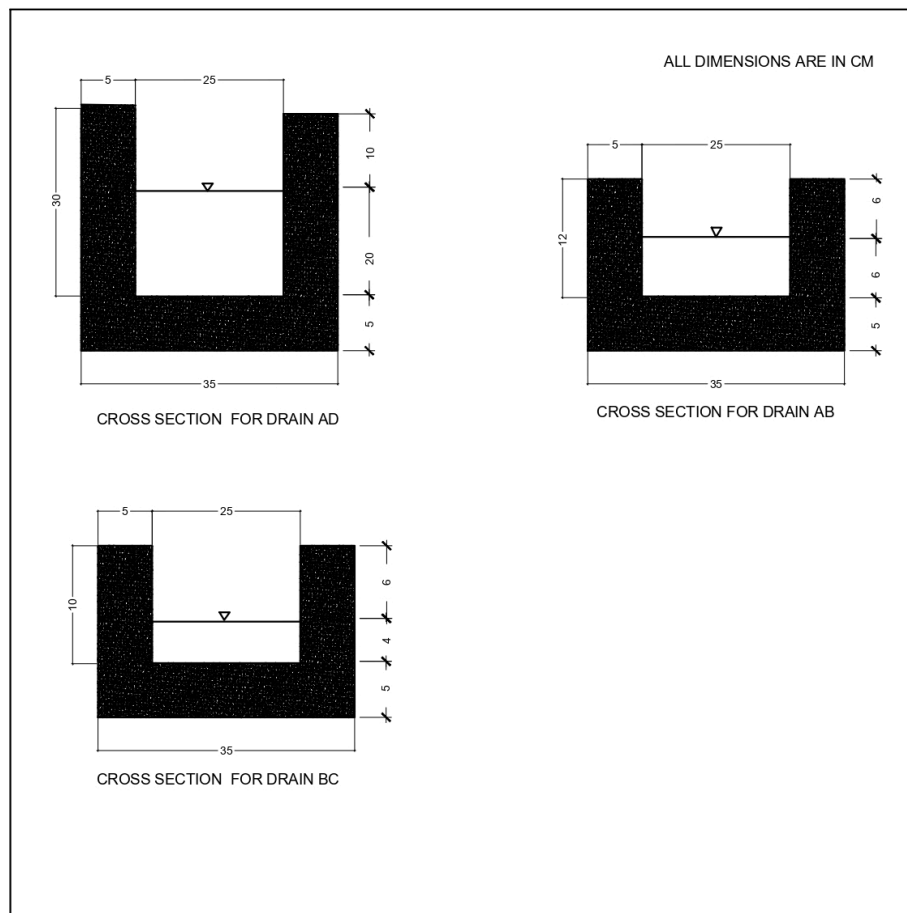
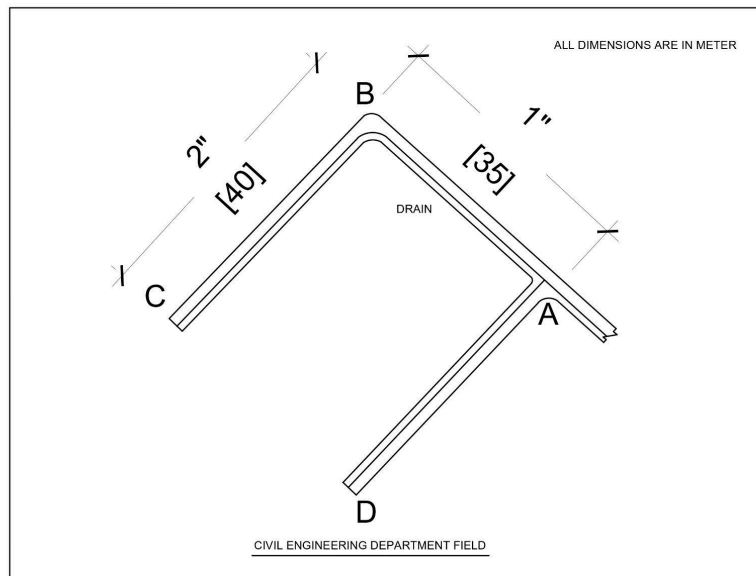
- Bed slopes: 1:1000 (vertical:horizontal)
- Depth:30cm

3) For BC drain

Bottom width: 25cm

- Bed slopes: 1:1000 (vertical:horizontal)
- Depth:10cm

6.DESIGN LAYOUT



7.COST ANALYSIS

We are using M15 grade concrete 1:2:4

For drain AD length of drainage 40 m

$$\begin{aligned}\text{Dry volume of concrete} &= ((50 \times 350) \times 2 + 250 \times 50) \times 40000 \\ &= 1.5 \times 10^9 \text{ mm}^3 \\ &= 1.5 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Wet volume of concrete} &= 1.5 \times 1.54 \\ &= 2.31 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of cement required} &= 1/7 \times 2.31 \\ &= 0.33 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of sand required} &= 2/7 \times 2.31 \\ &= 0.66 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of coarse aggregate required} &= 4/7 \times 2.31 \\ &= 1.32 \text{ m}^3\end{aligned}$$

$$\text{Cement} = 475.2 \text{ kg}$$

$$\text{Fine aggregate} = 0.66 \text{ m}^3$$

$$\text{coarse aggregate} = 1.32 \text{ m}^3$$

For drain **AB** length of drainage = 35 m

$$\begin{aligned}\text{Dry volume of concrete} &= ((170 \times 50) \times 2 + 250 \times 50) \times 35000 \\ &= 1.03 \times 10^9 \text{ mm}^3 \\ &= 1.03 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Wet volume of concrete} &= 1.5 \times 1.03 \\ &= 1.6 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of cement required} &= 1/7 \times 1.6 \\ &= 0.228 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of sand required} &= 2/7 \times 1.6 \\ &= 0.457 \text{ m}^3\end{aligned}$$

$$\text{Volume of coarse aggregate required} = 4/7 \times 1.6 = 0.914 \text{ m}^3$$

Cement = 328.32 kg

Fine aggregate=0.457m³

coarse aggregate=0.914m³

For drain **BC** length of drainage 40m

Dry volume of concrete=((50 x 150 x 2 + 250 x 50) x 40000

=1.1 X 10⁹ mm³

= 1.1 m³

Wet volume of concrete = 1.1 x 1.54

= 1.7 m³

Volume of cement required = 1/7 x 1.7

= 0.242 m³

Volume of sand required = 2/7 x 1.7

= 0.485 m³

Volume of coarse aggregate required = 4/7 x 1.7

= 0.971 m³

Cement = 348.48 kg

Fine aggregate = 0.485 m³

coarse aggregate = 0.971 m³

AD drain is bigger size we need reinforce for AD drain we are using

Fe 415 grade and 10 mm diameter in AD

Length of U bar,

n = 40000/900 = 44.45

Total length = 39996 mm = 40 m

Total length of longitudinal bar= 7 x 40000

= 280000 mm

= 280 m

Total length = 280 + 40 = 320 m

$$\begin{aligned}\text{Mass of steel} &= 0.6 \times 320 \\ &= 192\text{kg}\end{aligned}$$

Therefore cost analysis for material -

$$\text{Total cement} = 1152 \text{ kg}$$

$$\text{Number of bag required} = 23$$

For safety we will take 25 bag

$$\text{For that total rupees} = 20500 \text{ rupees}$$

$$\text{Total volume of fine aggregate} = 1.602 \text{ m}^3$$

$$\text{Total rupees for fine aggregate} = 5700 \text{ rupees}$$

$$\text{Total volume of coarse aggregate} = 3.205 \text{ m}^3$$

$$\text{Total rupees of coarse aggregate} = 10000 \text{ rupees}$$

$$\text{For Fe 415 steel total cost} = 11520 \text{ rupees}$$

$$\text{Therefore total material cost} = 47720 \text{ rupees}$$

8.Results and Analysis:

The open drainage system was designed to handle a total of 21 L/s of water. The final design involves:

3 drains,

Drain AB with a bottom width of 25 cm and depth of 12 cm, $Q = 3.15$ L/s.

Drain BC with a bottom width of 25 cm and depth of 10 cm, $Q = 2.1$ L/s

Drain AD with a bottom width of 25 cm and depth of 30 cm, $Q = 17.65$ L/s.

The results indicate that this configuration will prevent waterlogging in the field while maintaining soil strength.

9. Conclusions:

The design of the open-field drainage system for a 1400 m² area has been successfully completed. The system is capable of handling the expected water runoff from rainfall and preventing waterlogging. The chosen dimensions for the drains effectively address the drainage needs of the field.

10. Recommendations:

1. **System Maintenance:** Regular inspection of the drainage channels is recommended to ensure that they are free from blockages and sediment buildup.
2. **Monitoring:** The performance of the system should be monitored during the rainy season to identify any potential issues with water flow or insufficient drainage.
3. **Future Improvements:** Consider adding sediment traps at key points to prevent soil erosion and silt buildup in the drains.